Q2 2026 Post Earnings Analyst Call

Company Participants

- Manish Bhatia, Executive Vice President, Global Operations
- Mark Murphy, Executive Vice President and Chief Financial Officer
- Satya Kumar, Investor Relations
- Sumit Sadana, Executive Vice President and Chief Business Officer

Other Participants

- Aaron Rakers, Analyst, Wells Fargo
- Atif Malik, Analyst, Citi
- Chris Caso, Analyst, Wolfe Research
- Melissa Weathers, Analyst, Deutsche Bank
- Sam Feldman, Analyst, BNP Paribas
- Srini Pajjuri, Analyst, Royal Bank of Canada
- Vijay Rakesh, Analyst, Mizuho Securities

Presentation

Operator

Ladies and gentlemen, thank you for joining us and welcome to Micron's Post-Earnings Analyst Call. After the speaker presentation, we will host a question-and-answer session.

I will now hand the conference over to Satya Kumar, Investor Relations. Satya, please go ahead.

Satya Kumar {BIO 18241786 <GO>}

Yeah, thank you, and welcome to Micron Technologies' fiscal second quarter 2026 post-earnings analyst call. On the call with me today are Sumit Sadana, Micron's Chief Business Officer; Manish Bhatia, EVP of Global Operations; and Mark Murphy, our CFO.

As a reminder, the matters we're discussing today include forward-looking statements regarding market demand and supply, market trends and drivers, and our expected results and guidance, and other matters. These forward-looking statements are subject to risks and uncertainties that may cause actual results to differ materially from statements made today. We refer to the documents we have filed with the SEC, including our most recent Form 10-K and upcoming 10-Q for a discussion of risks that may affect our results.

Although we believe that the expectations reflected in the forward-looking statements are reasonable, we cannot guarantee future results, levels of activity, performance, and achievements. We are under no duty to update any of the forward-looking statements to confirm these statements to actual results.

Operator, we can now open the call up for Q&A.

Questions And Answers

Operator

We will now begin the question-and-answer session. Please limit yourself to one question and one follow-up. (Operator Instructions) Your first question comes from the line of Melissa

Weathers from Deutsche Bank. Your line is open. Please go ahead.

Q - Melissa Weathers {BIO 21061067 <GO>}

Hi, there. Thanks for letting me ask a question. I wanted to touch on the NAND side of things. So you guys are one of the only players to openly talk about Greenfield capacity adds so far. And then there are some industry participants saying that the growth in NAND bids going forward can be served by node upgrades alone without the need for capacity adds. So, can you talk about the decision to add Greenfield capacity there and what kind of trends you're seeing that gives you confidence to add that capacity?

A - Manish Bhatia {BIO 18045807 <GO>}

Hi, Melissa, it's Manish. I can -- let me talk a little bit about the decision, and then maybe Sumit can talk about the NAND demand trends. So we have said that in the past that while DRAM required greenfield wafer capacity, meaning new wafer start capacity growth to be able to meet the long-term demand trends. We saw NAND was able to meet that with technology transitions. And I think we still see that NAND technology transitions are able to provide strong bit growth going forward. And so our decision here, while, yes, reflecting confidence in the market demand outlook, which Sumit will talk about, was also driven by our continued space consumption for those technology transitions and for the next technology transitions that we'll be having in the future as well as our decision to locate more of our NAND R&D in the -- in Singapore where it's closer to our manufacturing.

And so, those two drivers of additional Clean Room space combined with our outlook on the market were really what was behind our -- in our confidence, frankly, in our product portfolio, which I'm sure Sumit will talk about, we're behind our decision to be able to break ground for our new NAND fab.

A - Sumit Sadana {BIO 7009258 <GO>}

And just on that, it's not a greenfield site, right? And I mean, you mentioned Greenfield in your question, and this is the existing site where we are adding this additional Clean Room space than we said.

So in terms of the demand side of the picture, I mean, we see very robust demand for NAND, driven by growth in the data center. AI servers are using huge amount of SSDs, high-capacity SSDs as well as high-performance SSDs. And this is working really well in our favor because our portfolio is doing exceptionally well. We are the first company in the world to have Gen6 SSD in the market, and this has been something that works well with NVIDIA systems. We have seen tremendous demand for it that we are not able to completely even come close to meeting. And then even high-capacity SSDs, we have seen tremendous growth and tremendous demand, and we have come out with a slew of new products that we are looking really excitedly to continue to grow our share in the data center SSD space from a record levels it reached at the end of last calendar year -- calendar 2025.

We have been growing to record shares in the data center every year for the last four years. So, we are very excited about that, and our supply is nowhere close to being able to meet the demand that we see for the foreseeable future. And so, this expansion that Manish mentioned is something we will put to good use to continue to grow our business with, obviously, in our focus on disciplined investing when it comes to our CapEx going forward.

A - Manish Bhatia {BIO 18045807 <GO>}

And I should just add that even though we're breaking ground now, that capacity -- that Clean Room isn't expected to provide -- new boost to our capacity until the second half of 2028. So we do think that not just for us, but across the industry in NAND Clean Room space over the medium term is still going to be a challenge for the industry, particularly as many in the industry have redirected some NAND Clean Room space towards DRAM.

Q - Melissa Weathers {BIO 21061067 <GO>}

Perfect. Thank you. And then back to the DRAM side, I wanted to ask pretty candidly on the pricing that we're all seeing. Next year, we have line-of-sight to a couple of DRAM projects coming online, including you guys bringing on a couple of those new fabs. So, how are you internally modeling the impact of that increase in supply going forward? Should we -- are you guys thinking pricing could come down next year or flatten out or decell? Just any color on how you guys are internally modeling the impact of all those new fabs coming online next year and the year after?

A - Manish Bhatia {BIO 18045807 <GO>}

And maybe -- I'll answer the supply side or what the industry -- and I think for the projects that we've talked about, which are the new Idaho Shell -- in the DRAM side, Idaho Shell as well as the -- our ramp in the Powerchip terminal facility that we've now acquired, we've said that the supply impact from those will be towards our fiscal '28 in terms of being able to meet revenue shipments. And I think that's sort of largely the case for the industry, in terms of new large Clean Room projects coming online. They will be really impacting later in '27 and really into '28 before you see meaningful supply, which is one of the reasons we have reiterated that we see the tight supply conditions to continue beyond '26.

A - Sumit Sadana {BIO 7009258 <GO>}

Yeah. I think we obviously don't provide any kind of price modeling or anything else type of discussion in that forward-looking way. But, we have mentioned that the demand forecast that we get from our customers for 2026, for 2027 continue to escalate, and despite efforts that we are making to increase the amount of supply we can bring online, and to a modest extent in '26, but some more in '27, is not really making that much of a meaningful dent in the gap. There is a lot of demand growth driven by AI, and also across different segments of the market, not just in the data center.

And we expect tight conditions for the foreseeable future and certainly beyond 2026, and not giving any further guidance beyond that.

A - Mark Murphy {BIO 16136048 <GO>}

I would just the last thing I would add, Melissa, is that, as Sumit mentioned, demand far exceeds supply, as Manish mentioned, yeah, these fabs talked about for us are coming on basically meaningfully in our fiscal '28. We always have the ability to modulate tool installs. So it's --, and that would be an option as well.

Q - Melissa Weathers {BIO 21061067 <GO>}

Perfect. Thank you, all three.

Operator

Your next question comes from the line of Aaron Rakers from Wells Fargo. Aaron, your line is open. Please go ahead.

Q - Aaron Rakers {BIO 6649630 <GO>}

Yeah, can you hear me?

A - Manish Bhatia {BIO 18045807 <GO>}

Yes.

Q - Aaron Rakers {BIO 6649630 <GO>}

Hello? Yeah. Sorry guys.

A - Mark Murphy {BIO 16136048 <GO>}

Yeah. We can hear you, Aaron.

Q - Aaron Rakers {BIO 6649630 <GO>}

Thanks for doing this. Perfect. Thanks, Mark. So thanks for doing this call. Two questions, one and a quick follow-up. I guess, Mark, maybe just to help model-wise, I'm curious if you could maybe frame the current quarter guidance between DRAM and NAND. What may be assumptions you're making in this tight environment around your ability to ship bits will that grow sequentially, and any kind of separation between the two buckets, between DRAM and NAND?

A - Mark Murphy {BIO 16136048 <GO>}

I can provide you. We don't break it out. I can provide you a little bit of color. I mean, let me just, maybe you heard the 2Q report out. So both DRAM and NAND pricing was up strongly, NAND, even more than DRAM, and both also grew volumes sequentially, NAND less than DRAM.

In Q3, we would expect price to again be the largest factor. But we've given you enough on CY26 industry bit shipments, and that those industry bit shipments are constrained by supply. And then our supply, we expect to grow in line with the industry. If you do all that, you can sort of determine that there's -- yeah, that you can assume some modest growth, volume growth in third quarter for both DRAM and NAND.

Q - Aaron Rakers {BIO 6649630 <GO>}

Okay. That's perfect. Thanks, Mark. And then, as a follow-up, maybe more architecturally in this environment, I'm curious, there's a tremendous amount of demand that you guys are seeing in servers. I think you've guided low-teens unit growth this year. But one of the architecture things that I'm starting to hear a little bit more about is the idea of CXL and memory pooling and whether or not some of these hyperscalers could drive better efficiency of their memory architecture.

I'm just curious at a high-level what your guys' thoughts are on that. Obviously, there's a lot of other factors, MRDIMMs, and just content growth in general, but any thoughts on CXL and having an effect on DRAM?

A - Sumit Sadana {BIO 7009258 <GO>}

Yeah, Aaron, CXL is likely to be experimented with at some of our customers, and they are different companies are in different phases of either assessing it or figuring out what they want to do with it, if anything. And we are certainly going to have our memory be available in those CXL configurations. Now my feeling is that the demand is in such a robust place in terms of gap between supply and demand being so significant that any and every available opportunity that can be used and deployed at scale is going to be something that customers will likely try if it is architecturally feasible and can be made to work with their software in their systems.

There is a lot of work to be done to bring these solutions to a large deployment scale, and it's not easy, and it's not something that can be pervasively used. There are lots of technical limitations as well. But I'm sure those experimentations will continue, and some limited deployments will be done in order to test it out and see how it performs.

Q - Aaron Rakers {BIO 6649630 <GO>}

Yeah. Thank you.

A - Sumit Sadana {BIO 7009258 <GO>}

Sure.

Operator

Your next question comes from the line of Jim Schneider from Goldman Sachs. Your line is open. Please go ahead.

A - Mark Murphy {BIO 16136048 <GO>}

Jim, can you hear us? You have a question? Operator, maybe go to the next.

Operator

It seems like we might be having some technical difficulties. (Operator Instructions). In the meantime, we will move on to Atif Malik from Citi. Your line is open. Please go ahead.

Q - Atif Malik {BIO 7312618 <GO>}

Thank you guys for hosting the call. On the call, you mentioned that the non-HBM margins are higher than HBM. So my question is really on allocation. How are you guys thinking about allocation? And the reason I'm asking that question is because one of your Korean peers has been reported to be raising pricing by 100%. And I know you had pricing mid-60s so DRAM, understand there's always delta across different memory makers. So are you leaving money on the table by not pivoting more towards non-HBM? So, just your understanding on HBM versus non-HBM allocation?

A - Sumit Sadana {BIO 7009258 <GO>}

Yeah, I mean, we definitely go through different quarters, and these relationships between different products can change over time. There is no doubt that HBM pricing was set for a large portion of the shipments in 2026 calendar year, late in calendar '25, as is generally the case with HBM, where the pricing is determined for some time before the start of the calendar year. And that kind of a model is a good model to have. It provides good stability and visibility into the business. The pricing that we negotiate has very robust ROI and profitability, and we feel good about that business.

Of course, the upsides that we have been able to create from our operations team and supply chain teams on the HBM, we have been able to sell those for even more robust levels of pricing as those upside volumes have materialized. With that said, we don't view a lot of these HBM versus non-HBM allocations to be that tactically focused. These are strategic things that we have to do in terms of providing our customers with match sets of products so they can build AI systems. You cannot ship DDR5 into AI system that doesn't have enough HBM in it, and vice versa.

So, there is a natural level of balance that is needed for the market to be appropriately having matched sets of products to be able to ship these at scale. So with that in mind, certainly, the HBM market margins are good, and the non-HBM margins, of course, have become even higher. But then this is not just a data center issue; the non-HBM margins outside the data center, meaning the DRAM margins outside the data center, are also exceptionally robust. So we don't tend to like just jerk around the allocations to different customers in different segments just based on where the pricing is. We have a goal of working with our customers to meet their business needs, and we do that with the -- with an intent of helping them meet their business goals as well.

Q - Atif Malik {BIO 7312618 <GO>}

Thanks, Sumit. And as my follow-up, NVIDIA talked about gross chips could be like 25% of the ultra-fast inferencing market. And my understanding is currently, these LPU chips have embedded SRAM at one of the Korean makers. And in the future, if these chips move to a Taiwanese foundry, will the SRAM be embedded on that chip, or can the SRAM be standalone and somehow bonded to the processor?

A - Sumit Sadana {BIO 7009258 <GO>}

There are definitely on-chip SRAM approaches that are currently in use, and there is, I've seen, some talk about bonded SRAM, et cetera. I would not want to comment on what directions our customers would go in terms of how they work with the SRAM over time. My main focus -- our main focus here is that -- we look at all of these systems as being very well balanced in the way they are evolving. So these SRAM-based systems complement in small uses the larger systems that are being utilized and deployed at scale, like the Vera Rubin, for example, and other similar systems that are based on ASIC accelerators.

And we see these as continuing to move the ball forward in terms of making the systems more balanced, more efficient, and the DRAM usage in these systems continues to grow over time and has gotten to levels, which, of course, we don't have adequate supply for. But over time, we continue to focus on the growth of these average capacities, the growth of the high-performance tiers, and even the growth of LP DRAM in these AI systems, all of which are really big positives for us on the DRAM side of the business.

Q - Atif Malik {BIO 7312618 <GO>}

Thank you.

Operator

Your next question comes from the line of Vijay Rakesh from Mizuho. Your line is open. Please go ahead.

Q - Vijay Rakesh {BIO 5884146 <GO>}

Yeah. Hi. Just a quick question on the NAND side. I know you mentioned that NAND is seeing a pretty strong uptake. Just wondering how to size it with the KV cash demand, what's the mix of just the -- AI data center, I should say, AI data center and NAND if you look out for calendar '26 versus last year, if you could -- if there is a way to kind of size that pickup in demand from KV cache, and I have a follow-up.

A - Sumit Sadana {BIO 7009258 <GO>}

When we made our prior forecast of the extent of growth we are likely to see in the data center space, it did include a view that KV cache would be a meaningful driver. And as that has become more in focus in terms of how the -- how our customers have been wanting to deploy their SSDs, definitely it has increased our view of the extent of demand coming from KV cache-related applications, and that has continued to cause our view of the total market opportunity to continue to grow.

I'll just remind us that something we have said the last time also, which is that we have also seen a significant uptick in demand for data center SSDs coming from shortages in HDDs. And we continue to see those shortages for the foreseeable future. That has been another driver. So when we put all of these together, the NAND market is significantly undersupplied to the demand in the data center, and that demand continues to escalate.

In part driven by KV cache, but also driven by just the insatiable appetite that these AI servers have to have fast storage capability available as these systems get deployed more and more. And so, the outlook is really strong. And, as we have mentioned earlier, our portfolio is incredibly well-positioned to continue to gain share in that space, including our KV cache applications by the way.

Q - Vijay Rakesh {BIO 5884146 <GO>}

Yeah. Right. Thanks. And just a follow-up on that. As you look at the CapEx, obviously, last couple of years, CapEx on NAND has been lighter. But as you look at the mix of CapEx for '26, '27 with the other CapEx numbers that went up, how would you look at the mix of DRAM to NAND CapEx, given both, DRAM demand is up, but you're also seeing a spike in ticket? Thanks.

A - Mark Murphy {BIO 16136048 <GO>}

Yeah, maybe I'll start. So Vijay, the CapEx is still going to be dominated by DRAM and HBM additions. Yeah, this FY26, we increased our outlook on CapEx to over $25 billion, which was up from the $20 billion we did on the last earnings call. Yeah, we -- the updated investment reflects this investment in Tongluo fab, which we communicated at February conference, and increases in US expansion. Again, it's DRAM and HBM-driven, including the increase. Today, we also provided more detail on construction, which we've said in the past was becoming a more material part of the build-out of, obviously, because we need Greenfield capacity. On the December call, we said expect that FY25-'26 construction would double. And so we expect FY26 construction to be mid to-high single-digit billions net when I talk CapEx, we're talking net CapEx.

Now, as we look out to '27, we did say that we project approximately a $10 billion incremental construction cost, and also for equipment spend to increase. Now, that '27 spend, NAND will begin to increase, but it will still be a much smaller portion of the spend compared to DRAM.

Q - Vijay Rakesh {BIO 5884146 <GO>}

Got it. Thanks, Mark.

Operator

Your next question comes from the line of Karl Ackerman from BNP Paribas. Please go ahead.

Q - Sam Feldman {BIO 24140642 <GO>}

Hi, can you hear me?

A - Manish Bhatia {BIO 18045807 <GO>}

Yes.

Q - Sam Feldman {BIO 24140642 <GO>}

Hi. Thanks for taking my question. This is Sam Feldman on for Karl Ackerman. So we've seen continuous HBM content uplifts with each new generation of XPUs, with a grown trade ratio of HBM, keeping the DRAM market tight and increasing AI memory requirements in the form of server DRAM, LP DRAM, and SRAM. Do you think HBM will continue to see such large content uplifts with each new generation of processors, or do you expect the content per XPU to eventually plateau?

A - Sumit Sadana {BIO 7009258 <GO>}

Well, we are not going to make long-term projections of where these average capacities will go because those are things that our customers are going to decide as their architectures evolve. What I will say is that if you look at the direction in which AI is trending and the types of things that are creating value in the AI domain for customers, they go towards more reasoning capability, and more longer context windows, and the ability to do more with agents and multi-agent orchestration. All of these things are really requiring more DRAM capacity and more DRAM bandwidth. And when it comes to delivering on that kind of bandwidth and being able to really optimize the system for all of the different stages of prefill and decode and different aspects of training as well as inference, you really come to the conclusion that these accelerators, whether it's GPUs or A6 do require increasing amount of HBM and increasing amount of DDR5 or LP5 capacity with time. And that's the trend that we have seen thus far, and you're also seeing, and we announced architectures that have been publicized.

And so, we feel certainly that the trend is -- has been clear. And not only has the trend been clear, but that the way our customers' customers, meaning the end customers, derive value out of the AI system and AI applications is very much connected and consistent with the trend of needing more DRAM capacity and bandwidth, which HBM delivers so effectively at very good efficient power consumption levels. And so that's the reason we have said in the past that memory is becoming a strategic asset in the AI era, is precisely one of the important drivers of that is precisely this trend that you really can't have a high-performing AI architecture or hardware infrastructure without all of those capabilities that DRAM and HBM bring to the table. So that's something we feel pretty good about in terms of where the market is headed.

Q - Sam Feldman {BIO 24140642 <GO>}

Great. Thank you.

Operator

Your next question comes from the line of Chris Caso from Wolfe Research. Your line is open. Please go-ahead.

Q - Chris Caso {BIO 4815032 <GO>}

Yes, thank you. Wondering if you care to update your view of long-term bit growth for both DRAM and NAND. And I know you mentioned in the call low-20s in DRAM 20% for NAND, but those are obviously supply-constrained numbers. And maybe you speak to the extent to which you think long-term bit growth is increasing as a result of all the things that we've been talking about.

A - Sumit Sadana {BIO 7009258 <GO>}

Yeah. I mean, we haven't provided a new long-term bit growth number. I think you're you have seen that in the past, we have spoken about high-teens, mid to high-teens type of ranges for DRAM. And yet you have seen that last year, this year, our forecasts have been more robust than those levels. And we continue to feel like we are in an extended space of robust industry demand that, obviously, due to HBM being part of these numbers, with its trade ratio, is just stressing the entire industries and certainly our capabilities to be able to meet those demand numbers. So you're right, I mean these numbers, at least in the foreseeable future, are all supply-limited numbers rather than the actual level, true level of demand.

So yeah, I mean that's sort of the environment we are in. We do expect that next year, again, we will have a fairly robust level of growth in calendar '27. But yeah, we are not providing a long-term number beyond that commentary.

Q - Chris Caso {BIO 4815032 <GO>}

Got it. Thank you. As a follow-up, obviously, there's been a lot of discussion about clean room constraints. And as you folks have pointed out, you're starting greenfield in Singapore, and it's not available till the end of '28. I guess you probably can't speak for the industry, but for at least for Micron, at what point do you think you can get caught up with having enough clean room capacity that gives you some headroom for what the customers need?

And then obviously, then it's going to take time to move the tools into that. But it goes to the sustainability of what's going on right now and the extent to which the clean room constraints are going to drive that sustainability?

A - Manish Bhatia {BIO 18045807 <GO>}

Well, I think I think, Chris, the first part of the question is really around what Sumit had described in terms of the long-term demand, and we're continuing to evaluate all the different demand drivers and signals, including what all the announcements from GTC this week. In terms of the availability of space for us and the industry, certainly we have -- we see this relative constraint for all the major DRAM players to be there through this year and into next year, with major meaningful improvement to clean room space availability only out into '28.

But of course, as we think through our projects that we've announced go beyond that into 2028, '29, '30, as we think -- as we talked about those projects, the timeline for us to get to the point, as you mentioned, will be a function of the demand. And we will be nimble to be able to adjust equipment, orders, and equipment installations in order to stay aligned with whatever the demand is as it plays out over that time horizon.

A - Sumit Sadana {BIO 7009258 <GO>}

I'll just add to what Manish said, which is that as we look at the demand side of the picture and we have been engaged with our customers, several of them to be talking about these five-year, multiyear SCA agreements. And in the context of that, of course, we are assessing their longer-term demand, their five-year demand, for example. And we are assessing those against our own supply capabilities.

And within that timeframe, we are also seeing the emergence and growth of some really exciting new demand vectors, including things like robotics, which we expect to become a very major demand driver. And so when we put all of those things in the equation, we don't have a high confidence view yet as to when the supply will be able to catch up with demand because the escalation of demand from these various vectors is just very phenomenal. And so you know, to answer the question as to when do we have a high confidence view that the supply will be able to catch up with demand, we don't really have a high confidence view yet as to when that would happen.

Q - Chris Caso {BIO 4815032 <GO>}

Thank you.

Operator

Your next question comes from the line of Srini Pajjuri from Royal Bank of Canada. Please go ahead.

Q - Srini Pajjuri {BIO 21216060 <GO>}

Thank you. Mark. I want to ask the previous question slightly differently on CapEx. You talked about CapEx being for the construction being high single-digits in '25, growing by $10 billion. So that suggests that roughly next year, half-and-half between construction and equipment. For the projects that you already announced, do you think '27 is the peak year for construction CapEx? If so, what is the normalized, I guess, mix between construction equipment through the cycles that you kind of think about?

A - Mark Murphy {BIO 16136048 <GO>}

Srini, we're not going to give any more breakdown than we've provided. Yeah, it will be lumpy. Just to be clear, the $10 billion was in reference to '26 to '27. It wasn't quite -- you may have said that, but I just want to make sure it's clear. And we're, of course, going to modulate spend as we see demand and to maintain stable bit share. As Sumit mentioned, it's not clear if we're investing as we can, including this recent acquisition of the Tongluo fab to put capacity in, but it takes time and a lot of effort; we can't get meaningful bits until our fiscal '28.

So, beyond '27, of course, we're going to be very disciplined, and '27 could be a CapEx could go down after '27, but we're not making that call at this point. We're just investing through this next several-year period that we can try and get the supply we need for our customers.

Q - Srini Pajjuri {BIO 21216060 <GO>}

Okay. Then maybe a follow-up. For the next few quarters, there were questions on gross margins. Obviously, you don't want to comment on pricing, but could you give us an idea how to think about any puts and takes on the cost side, and also any mix dynamics that we should be aware of? And also, given the higher CapEx, how should we kind of think about depreciation over the next few quarters and any start-up costs that you anticipate? Thank you.

A - Mark Murphy {BIO 16136048 <GO>}

Yeah. So good questions. Maybe tackle first, just on costs. We continue to execute really well on cost reductions. Manish can comment some more. But as we've talked about FY26, we have the benefit of the node transitions, 1-gamma, in particular, on DRAM and G9 on NAND that are driving our bed growth. So we get a lot of efficiency there and cost downs, and the 1 gamma, for example, replacing one alpha capacity. So, that's good, and the spend control has been outstanding. So, as busy as we are, as good as the numbers are, the discipline in the business is very good. Including managing, by the way, all this geopolitical, and the team has really been on top of that. There's no impact to our operations. So that's good.

You know, I think -- to your question on startup costs, as we bring on ID1 and then there's going to be somewhat Tongluo. And, yea, I made some comments, maybe it was a year, year and a half ago, talking about startup costs, and those comments are still applicable. And at the time, I said I think between a point, two points of costs.

Now, revenue was much lower at the time. So if we dollarize that, it's probably $100 million -- $100 million to $200 million per quarter starting in the next quarter or so, and then continuing on through '27, and then it would come down off of that. So again, at these revenue margin levels, it's a much smaller impact, 50 basis points or less. And then, I think, was there -- what was the third question?

Q - Srini Pajjuri {BIO 21216060 <GO>}

Yeah, depreciation around -- (Multiple Speakers)

A - Mark Murphy {BIO 16136048 <GO>}

Oh. Yeah, depreciation just depends on when production wafers are out and then the useful life of those assets when they're put in service. And keep in mind, it's -- that because this is greenfield, this is a very long depreciation life. So that's important to keep in mind.

A - Manish Bhatia {BIO 18045807 <GO>}

So just a couple of things I'll add, Mark, ask you had some comments on cost. I think you covered a lot of them. 1-gamma is going really well, Gen9 are both going well. We had positive comments on those. The only other one I'll add is HBM. I think our HBM3E 12-Hi has continued to execute well as we've gone through the last two -- gotten to high volume over the last couple of quarters. And we actually see HBM4 even though we're in the early stages of the ramp, having an even faster yield ramp than HBM3E 12-Hi. So our HBM cost structure, we feel very good about both HBM3E as well as HBM4, and providing improvements for us over previous periods.

And then the other comment I'll just have, I think, Mark referred to geopolitical and managing, I think, the costs relative to geopolitical issues. I think you meant about the Middle East and the sort of the media reports regarding various input disruptions, but we don't see any supply risks and very, very minimal impact to cost at this time.

A - Mark Murphy {BIO 16136048 <GO>}

Yeah, maybe -- just maybe one housekeeping issue since we're covering sort of those issues now. On OpEx, we indicated that our OpEx would be ticking up on -- due to R&D costs. So in the fourth quarter, we would expect OpEx to be closer to $1.6 billion, just given that with part of it's the extra week, or part of it is just again this increase in R&D spend, which we think is completely appropriate to drive the technology, the increased value of memory, Micron's position, and then just customer demand on specific projects. And then we would expect in '27 for that OpEx number to be over 1.6, probably kind of a 1.7 run rate number, and stabilize from there.

Q - Srini Pajjuri {BIO 21216060 <GO>}

Thanks, Mark.

Operator

This concludes today's call. Thank you for attending. You may now disconnect.